

Dancing through the Decades:

A Study of Musical Trends from 1921 to 2020

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Abstract

This study investigates the key predictors that could help determine the era of a song on Spotify. Using a random forest model, we classified songs into four distinct eras: 1920-40s, 1950-60s, 1970-90s, and 2000-10s. Our analysis indicated that acousticness, danceability, energy, instrumentalness, valence, tempo, and loudness were the most significant features in predicting the era of a song. Variables such as key and liveness proved to be less useful for classification. The final random forest model achieved an accuracy of 70.22% with a kappa statistic of 0.5931. The strength of our classification model also varied across different eras.

I. Introduction

Do you ever scroll through Spotify playlists and suddenly find yourself in an '80s mood? Why does 2000s music feel so nostalgic to so many people? What makes '90s music distinct from other decades?

This report explores four key research questions:

- 1) Can we predict the era in which a song was released based on different musical metrics?
- 2) Which musical metrics are most indicative of the era in which a song was released?
- 3) For which eras are musical metrics most similar?
- 4) Does the uniformity of the musical metrics of same-era songs change throughout time? In which era are the songs most uniform or diverse?

II. Background

The dataset used in this study was provided by Kaggle and consisted of over 160,000 songs from 1920 to 2020 collected using Spotify's API. It contained 19 features that were relevant in analyzing characteristics and music trends through the decades, with some of these being words or metrics created by Spotify (e.g., instrumentalness, speechiness, and acousticness). We explored how some of these features correlate with the era of release and focused on how they change over time.

Variable	Description	Values
Acousticness	How acoustic a song is	1, 0
Danceability	How well a track is suited for dancing	1, 0
Energy	Intensity and activity of a song	1, 0
Duration	Total song length (ms)	
Instrumentalness	How instrumental (few vocals) a song is	1, 0

Valence	How musically positive a song is	1, 0
Tempo	The pace of the song (BPM)	
Liveness	Detects presence of a live audience	1, 0
Loudness	Average volume of the track (dB)	
Speechiness	Presence of spoken words in a song	1, 0
Key	Musical key of the song (C to B in ascending order by pitch)	0-11
Year	Year of track	1921-2020

Figure 1: Description of Variables

III. Exploratory Data Analysis

We conducted data preprocessing by grouping the “year” variable into four distinct eras: the 1920-40s which was dominated by jazz, swing, and big band; the 1950-60s which was dominated by pop, doo-wop, and rock and roll; the 1970-90s which was dominated by disco, funk, dance, and hip-hop; and the 2000-10s which was dominated by pop and hip-hop.

We created boxplots with normalized features to compare the differences in variation between eras and hypothesize which features would contribute most to the classification model. Features such as Duration_ms, Valence, Tempo, Liveness, and Key showed a relative lack of variation between eras, indicating that they would likely be less useful in classifying the release era in which a song was released.

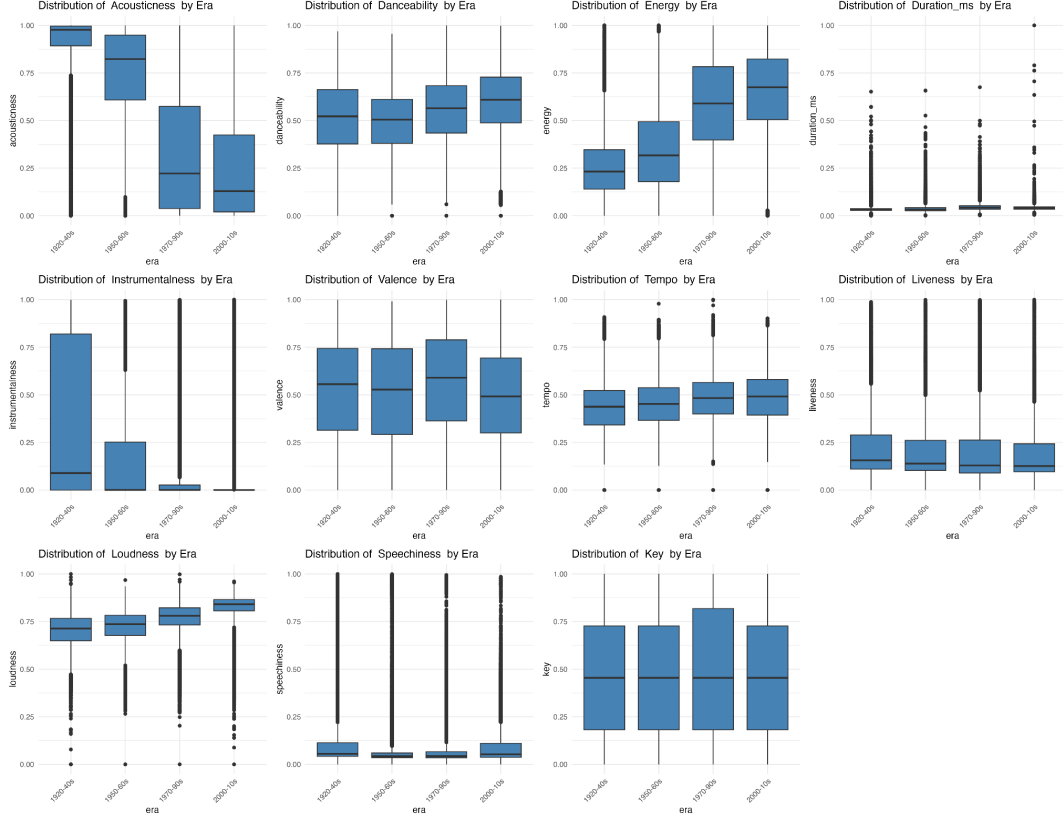


Figure 2: Boxplots of Normalized Features

We also created a standard deviation table with unnormalized features to interpret dispersion in the same units as the data and analyze the natural variability of each feature.

era	acousticness	danceability	energy	duration_ms	instrumentalness	valence	tempo	liveness	loudness	speechiness	key
1920-40s	0.2415476	0.1775765	0.1679244	151921.98	0.3908347	0.2661394	31.23720	0.1641324	5.541730	0.2675273	3.480322
1950-60s	0.2679458	0.1611221	0.2183729	138243.51	0.3374951	0.2699224	30.58347	0.1730616	5.299671	0.1346981	3.449700
1970-90s	0.3110209	0.1725520	0.2460486	107715.04	0.2529512	0.2594586	29.44433	0.1951264	4.652581	0.0829808	3.530223
2000-10s	0.2917353	0.1734434	0.2254571	90312.06	0.2204621	0.2498550	30.88552	0.1589350	4.524076	0.1046053	3.575972

Figure 3: Standard Deviation of Features by Era

IV. Statistical Model

Due to its flexibility, robustness, and lower overfitting risk, a Random Forest model was chosen to classify the data. The dataset was randomly split into training data and testing data, which consisted of 80% and 20% of the dataset, respectively. We normalized the numerical features so that features with large numerical ranges such as duration or loudness would not dominate the model. Model performance was evaluated using accuracy and a confusion matrix, while feature importance was evaluated using impurity reduction.

V. Results

The Random Forest model had a relatively high capability in successfully classifying the era of release based on the musical metrics, achieving an accuracy of 70.2%. Acousticness and loudness were the most important features for predicting the release era of a song.

Prediction	1920-40s	1950-60s	1970-90s	2000-10s
1920-40s	4150	1121	197	97
1950-60s	1209	5305	1588	578
1970-90s	209	1410	8610	1888
2000-10s	60	220	1513	5827

Figure 4: Confusion Matrix

Metric	Value
Accuracy	0.7031
Kappa	0.5944
P-Value (Acc > NIR)	0.0000
McNemar's Test P-Value	0.0000

Figure 5: Overall Model Statistics

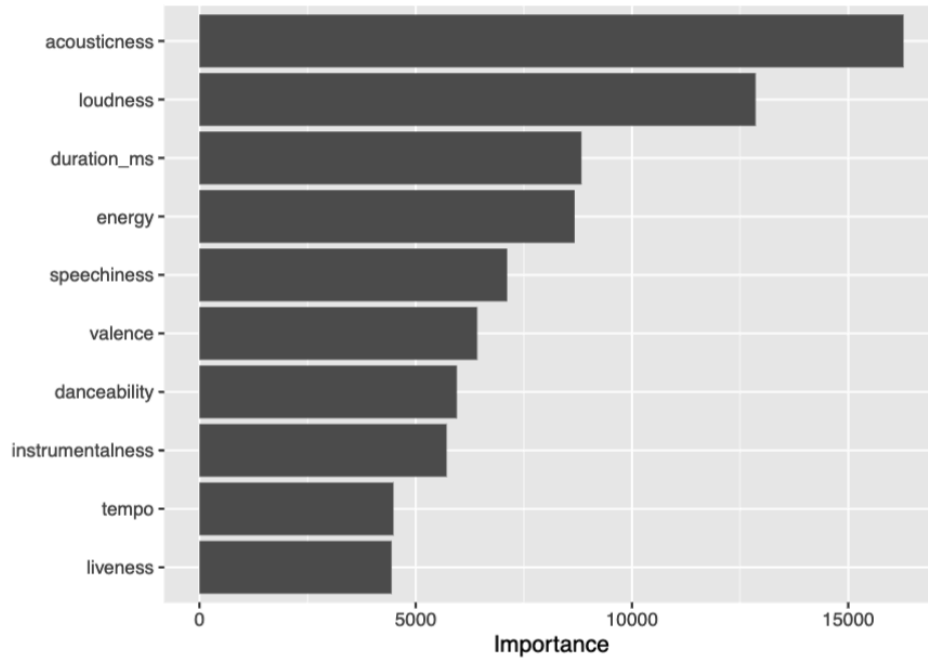


Figure 6: Feature Importance Plot

VI. Conclusion

While there was not a significant change in uniformity over time, variations in the standard deviations of features showed that certain eras had more individually diverse and uniform musical metrics than others. For example, the 1920-40s had the highest standard deviation for song duration and tempo but the lowest for acousticness; this can be explained by the fact that this era occurred before digital streaming and algorithms shaped the industry standard for song length and format and new genres popularized synthetic sounds over acoustic ones (as expected, the 2000-10s had the lowest standard deviation for duration and the second lowest for acousticness). These results show that cultural trends and technological advancements strongly influenced the music styles of different eras, leading to the observed similarities and differences.

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